

A comprehensive review on the fabrication of cassava peel-derived polysaccharides and their biocomposites for sustainable food packaging applications

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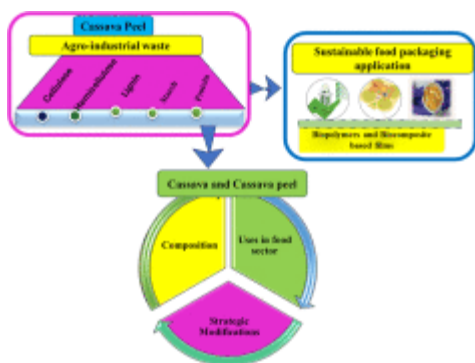
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In the past decades, agricultural waste-derived sustainable packaging has received considerable attention in the research and development field. The key motivating factor for agro-waste-derived packaging is to mitigate the problems caused by fossil-based packaging, which creates negative impacts on the environment. Nowadays, cassava peel, an agricultural waste material, has been extensively used as a natural source for the development of polysaccharides such as starch, cellulose and nanocellulose. These polysaccharides can be used as a matrix or reinforcement in the design and development of stringent food packaging materials. Moreover, cassava peel has become one of the most utilized agricultural wastes in waste management, with potential benefits in cassava processing industries. Thus, this review addresses the development of sustainable packaging materials from cassava peel-derived biopolymers. This review also discusses the methods used for the extraction of cellulose, nanocellulose and starch from cassava peel and some of the properties associated with these biopolymers. In addition, the thermal, mechanical and barrier properties of various cassava peel-based composite films and their application in food packaging are summarized. Finally, the safety aspects and future outlook of cassava peel-derived biopolymers for sustainable development are reviewed.

In the past decades, agricultural waste-derived sustainable packaging has received considerable attention in the research and development field.



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Sustainability spotlight

Cassava (*Manihot esculenta*) is recognized for its versatility and wide consumption across various regions globally. From cassava processing industries, a huge amount of cassava peels is generated as agro-waste. These peels are valued for their rich content of starch, dietary fibre (cellulose and hemicellulose) and lignin, which can be utilized in various ways to reduce waste and provide additional economic and environmental benefits. This review primarily focuses on the extraction of cellulose, nanocellulose and starch from cassava peel. The review also provides a comprehensive overview of the thermal, mechanical and barrier properties of various cassava peel-derived biopolymer-based composite films, which are utilized in various packaging applications. The approach not only supports waste valorization but also contributes to environmental sustainability by replacing the conventional synthetic packaging materials. Thus, the valorization of cassava peel into high-performance biopolymers presents a promising pathway towards eco-friendly and innovative packaging solutions that harmonize technological advancement with environmental sustainability.

Introduction

In the current trend of sustainable development, environmentally benign materials have established a momentous interest in the 21st century due to the rising global population, rapid urbanization, and the desire for higher standards of living. This has emerged as the central focus of the Sustainable Development Goals (SDGs)¹ as it addresses three of the

most critical challenges that our world is facing today: sustainable consumption and production (SDG Goal 12), the fight against climate change (SDG Goal 13), and the preservation of marine life (SDG Goal 14). The design and development of sustainable packaging systems has also received a considerable attention in the past years.² Additionally, the fossil-derived plastics have created an uncomfortable conundrum to the society with increased solid waste and carbon footprint, harming the society over the past few decades.³ Over the years, the issue of plastic waste has emerged as a significant concern, primarily due to the excessive consumption of plastics and their resistance to decomposition and substantial contribution to landfills and water pollution.⁴

In the early 1800s, folding cartons were manufactured using paperboard, and in the 1850 s, corrugated boxes were developed, which were one of the recommended shipping containers.⁵ Various plastics were discovered in the 1800 s, such as vinyl chloride, styrene, and cellulose nitrate, which were later explored in the development of food packaging.⁵ Among the available packaging materials, plastic has been the most fascinating one since the very early days, which created a huge amount of solid wastes. In 2015, approximately half of plastic waste originated from single-use plastics, which are often used in packaging and are designed for a one-time use before being discarded.⁶ This fossil-derived packaging material has caused substantial environmental pollution by generating greenhouse gas emissions during its production and disposal while also consuming finite fossil resources. Food packaging alone represents about 50% of plastics derived from fossil fuels. Once discarded, such plastics persist in the environment for decades, gradually fragmenting into microplastics that can seep into the ecosystems and enter the food chain, leading to bioaccumulation. An effective approach to address this issue is the replacement of fossil-derived polymer materials with biodegradable/renewable alternatives.⁷ This approach aims to facilitate the decomposition of biodegradable materials with the aid of microorganisms, leading to a reduction in ecological impacts.⁴

Various biomass obtained from agricultural residues, forestry residues, and urban miscellaneous waste are extensively valorised for sustainable food applications. As represented in [Fig. 1](#), the agricultural residues include farm-based, agro-industrial-based and others. Farm-based waste is mainly generated at the time of crop cultivation, which includes straw, husks and stalks. However, the agricultural wastes produced from the cultivation and processing of agricultural products are potential sources of polysaccharides, including starch, pectin, cellulose and nanocellulose. These waste streams are widely used as raw materials for the development of biopolymers. Moreover, agro-industrial waste materials, including sugarcane bagasse, bamboo bagasse, mango bagasse, apple pomace, coffee husk or pulp, and cassava, are extensively utilized for various applications.^{8, 9} These agricultural byproducts can also be transformed into

biofiber and biopolymer materials, which can be employed for versatile food packaging applications, such as bioplastic packaging, disposable packaging, trays and food coatings, as well as in long-lasting uses like plastic mulch, pharmaceuticals, medicinal products, and automotive components. Apart from this, forestry residues consist of logging wastes, including branches, bark and treetops, as well as wood-processing byproducts such as sawdust and offcuts. Furthermore, urban miscellaneous waste includes municipal solid waste from households and cities, livestock and poultry manure and other organic discards.



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Fig. 1. Various sources of biomass with potential in sustainable development.

Among these diverse residues, one of the important agricultural wastes currently used for industrial purposes is cassava peel. Cassava (*Manihot esculenta*) is a versatile and widely consumed root vegetable found in various regions globally. In India, it is mostly cultivated in the southern states, such as Tamil Nadu, Kerala, and Andhra Pradesh, and the north-eastern states such as Assam and Meghalaya. Cassava is well known for its starchy tuber, which can be used in various food applications. It is primarily valued for its starch content, as it contains around 60–70% of starch on a dry basis. It also contains lignocellulosic components, including cellulose, hemicellulose, and lignin.¹⁰ The production of cassava flour inherently generates by-products such as cassava wastewater (liquid waste) and cassava peels (solid waste). Cassava wastewater is obtained in the drying steps and contains a high percentage of organic compounds. On the other hand, cassava peel is a lignocellulosic residue,¹¹ comprising approximately 45% starch, 27% hemicellulose, 14% cellulose, 11% lignin, 10% crude fiber, and 3.5% protein. Moreover, the use of cassava peel in sustainable food packaging is an innovative and eco-friendly approach to reducing waste and promoting sustainability in the food industry. It can be used for preparing

biodegradable films, bioplastics as well as biocomposites. Cassava peel extracts can be used as a natural preservative in food packaging. The antimicrobial properties of cassava peel can also help to prolong the shelf life of food products. Beyond the food sector, cassava peel can also be utilized as an organic fertilizer to enhance soil fertility. Recent studies also highlighted the innovative approach of cassava peel in the field of the acoustic material industry for the development of acoustic panels. Additionally, cassava peel has been explored in microbial fuel cells, where it serves as a supplementary energy source, contributing to renewable energy generation and improved energy efficiency. In the field of environmental management, cassava peel serves as a sustainable coagulant aid in wastewater treatment. Collectively, these diverse applications emphasize the role of cassava peel valorization in advancing sustainability, resource recovery and bioeconomy practice.

Polysaccharides synthesized from cassava peel represent a sustainable alternative to synthetic polymers. Cassava peel-based polysaccharides, such as cellulose, starch, and nanocellulose, can be extracted using chemical, mechanical and chemo-mechanical treatments. The extracted polysaccharides are utilized to develop biocomposites, in combination with other polysaccharides, for food packaging applications. Cassava peel-based polysaccharides can be processed into edible films and coatings that are used to extend the freshness of food products. These films can serve as barriers to moisture, oxygen, and other contaminants, helping to preserve the quality and freshness of perishable foods. They can also be used as thickening, stabilizing and gelling agents in various food products, as represented in [Fig. 2](#). The cellulose fraction can be converted into microcrystalline cellulose, which functions as a stabilizer and is extensively used in the pharmaceutical industry. Its nanoscale derivatives (nanocellulose) exhibit superior mechanical and barrier properties, enabling their utilization in membrane fabrication for polymer electrolytes, food packaging, pharmaceutical carriers, and cosmetic formulations. Starch isolated from cassava peel has been utilized for the synthesis of polyhydroxyalkanoates (PHA), preparation of edible coatings, development of biodegradable plastics, and even the design of energy-storage devices such as supercapacitors. Similarly, pectin derived from cassava peel serves as an effective edible coating, particularly for prolonging the shelf life of fresh fruits. Lignin, another important component, is employed in char production and as a biodegradable filler in polymeric composites. Hemicellulose, on the other hand, can be valorized into biofuels, sustainable packaging materials, and functional adsorbents for environmental applications. Moreover, cassava peel-based polysaccharides can be employed to encapsulate and protect sensitive food ingredients or bioactive compounds. Based on the discussion, this review demonstrates a strategic roadmap, highlighting the importance of cassava and the

fabrication of cassava peel-derived polysaccharides and biocomposites for sustainable food packaging applications.



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Fig. 2. Various cassava peel-derived materials and their uses.

Composition and uses of cassava and cassava peel

Composition and uses of cassava

The chemical composition of cassava may differ based on various factors, including its variety, the composition of the soil, environmental conditions, geographic location, and the selected part of the plant. In cassava, starch is found in high concentrations, while protein, fat, fibres, and minerals are found in low concentrations. Additionally, the higher carbohydrate content in fresh cassava root makes it a good energy source.¹² Around 64–72% of the carbohydrate is found in the form of starch in cassava roots, and the remaining portion includes sucrose, fructose, glucose, and maltose.^{13, 14} Cassava roots consist of 32–35% carbohydrates on a fresh weight basis. In sweet cassava, the sucrose content is found to be more than that in other varieties.¹⁴ The majority of starch in cassava is stored in the amyloplast region, and the edible starch content in cassava roots can vary, depending on several factors including the cassava variety, growing conditions, as well as the maturity of the roots. It ranges from 80% to 90% of the total weight of the root. In the case of cassava leaves, the content of amylose can vary within a range of 19% to 24%. The fiber content of cassava leaves also varies based on the variety and age of the cassava roots. In the case of fresh cassava roots, the fiber content is less than 1.5%. Moreover, the protein content of cassava root is relatively lower as compared to other starchy foods, typically around 1–3% on dry weight basis. Cassava contains a relatively low amount of fat, typically ranging from 0.1% to 0.3%. Although cassava has low overall fat content, it

consists of 45% nonpolar lipids and 52% different glycolipids, with galactose-diglycerides being predominant.¹² Cassava plants indeed offer a variety of essential nutrients, making them a valuable dietary resource, particularly in regions where they are used as a staple food for millions of people. Cassava roots are used in various ways across the globe due to their versatility and nutritional value. Grating cassava roots and pounding them into a pulp to make porridge is one of the traditional methods of preparing cassava in some regions.¹⁵ Mingao and manicuera are two traditional Amazonian food products that are prepared from cassava. Fermented cassava is used for the preparation of mingao, whereas manicuera is prepared by using boiled cassava juice. Dumby is another food product that is prepared from boiled cassava root.¹⁶ Fufu is one of the traditional cassava-derived fermented foods of West Africa, which is prepared by crushing and sieving the fermented cassava and then cooking it in water.¹⁶ Apart from this, some other food products that are obtained from cassava are Gari, Lafun, Kpokpogari, Purupuru, and Peujeum.

Cassava has a wide range of industrial applications beyond its traditional use. It is widely used as ingredient in bakery products, extruded snacks, cereals and in biodegradable packaging materials, such as bags, containers, and utensils. There is a growing demand for alternative flours in the bakery industry due to various factors such as nutritional considerations, gluten intolerance, and wheat allergies. Cassava flour is indeed one of the options that have gained popularity as a substitute for wheat flour, especially in gluten-free and functional bakery products. Casabe, made from cassava flour, is a traditional type of bread in South America. It has a unique taste and texture, which can vary depending on the preparation method and regional variations.¹⁷ One of the studies conducted by Chinma and Gernah in 2007 focused on the preparation of cookies using a composite of cassava flour, soybean, and moderately ripe mango fruits. The researchers created cookies with different proportions of these ingredients, ranging from 100% cassava flour to various combinations of cassava, soybean, and mango. They also used 100% wheat flour cookies as a control group for comparison. The study explored the use of a composite of cassava flour, soybean, and mango to create cookies with improved nutritional content.¹⁸ The application of cassava starch in the production of edible food packaging has gained attention due to its biodegradability and versatility. In 2017, Piñeros-Hernandez incorporated polyphenol-rich rosemary extracts into cassava starch films, leading to the development of active food packaging materials with antioxidant properties. The incorporation of rosemary extracts preserved the quality of the food and also maintained the biodegradability of the packaging materials.¹⁹ In Thailand, plastic films have been produced from 100% cassava starch through a process called “annealing”. These cassava starch-based films are extremely biodegradable, offering an eco-friendly alternative to traditional plastic films.²⁰ Further, cereal starch can also be used in the formation of sugar

syrups that can be used in various food and beverage applications.²¹ Glucose syrups obtained from cassava starch can be used as sweeteners in the production of candies. Additionally, in the pharmaceutical and detergent industry, cassava starch is used in different forms.²²

Considering the rising demand for cassava peel and cassava-based products in the food processing and packaging industry, it has been observed that this crop will maintain its significant contribution towards ensuring food security across the world. Cassava varieties cultivated across the world contain a natural toxic compound known as cyanogenic glycosides.²³ Cassava varieties are often described as being bitter or sweet by reference to the taste of fresh roots, and this partly correlates with cyanogen concentrations.^{24, 25} Bitter varieties are associated with high concentrations of cyanogenic glycosides ($>100 \text{ mg kg}^{-1}$ fresh weight), which causes various environmental issues such as drought, low soil fertility, pest attack, and others.²⁴ Sweet varieties have a high concentration of free sugars and a low concentration of toxin content. Along with raw cassava, cassava peel also contains cyanogenic glycosides, particularly linamarin and lotaustralin, which are converted into poisonous hydrogen cyanide by the enzyme linamarase. Owing to this issue, any portion of cassava in its raw form cannot be consumed and requires proper detoxification methods before consumption. Common methods such as drying, grating, fermentation, boiling, steaming, and others are used for the detoxification of cassava. Apart from the toxicological aspects, cassava also contains antinutritional compounds such as phytic acid, tannins, oxalate, nitrates and saponins. Phytic acid has a strong ability to chelate minerals like iron, zinc, and calcium, and thereby reduce their bioavailability. Nitrates, on the other hand, may lead to methemoglobinemia, whereas saponins disrupt cell membranes, affecting nutrient absorption. Therefore, prior to the utilization of cassava or its peel as a food product or in packaging applications, these issues must be carefully addressed, and proper processing strategies need to be adopted for its safe and effective use.

Compositions and uses of cassava peel

Cassava peel is considered an important byproduct in cassava-based processes, which is utilized in various applications. It contains a significant amount of starch, dietary fibre (cellulose and hemicellulose), and lignin. The percentage composition of lignocellulosic biomass in cassava peel, as reported by various researchers, is represented in [Table 1](#). Moreover, the protein content in cassava peel is generally low compared to cellulose and starch, and it is approximately 3.5–5.29%.²⁶ Cassava peel also contains 3–7% of ash, which includes various minerals such as calcium, potassium, and magnesium. Cassava peel also contains a significant amount of cyanogenic glycosides, with levels around

710.98 mg hydrogen cyanide (HCN)/kg in unprocessed peel meal.[27](#) Nowadays, cassava peel-based polysaccharides are widely used as a reinforcing agent in various packaging industries, so it is very essential to reduce the cyanide level from cassava peel, otherwise it may cause cyanide poisoning. In this regards, proper soaking, drying and scraping can reduce the content of cyanide up to an acceptable level. Fermentation is also one of the techniques that is used to reduce the cyanide level to a safe level. In the fermentation process, microorganisms are used to convert the cyanogenic glycosides into safer compounds like cyanohydrins, subsequently breaking them down into harmless byproducts such as ammonia.[28](#) The Food and Agriculture Organization (FAO)/World Health Organization (WHO) (1991) recommended the limit value for the safe consumption of cassava products as 10 mg HCN/kg.

Table 1. Composition of cellulose, hemicellulose and lignin in cassava peel

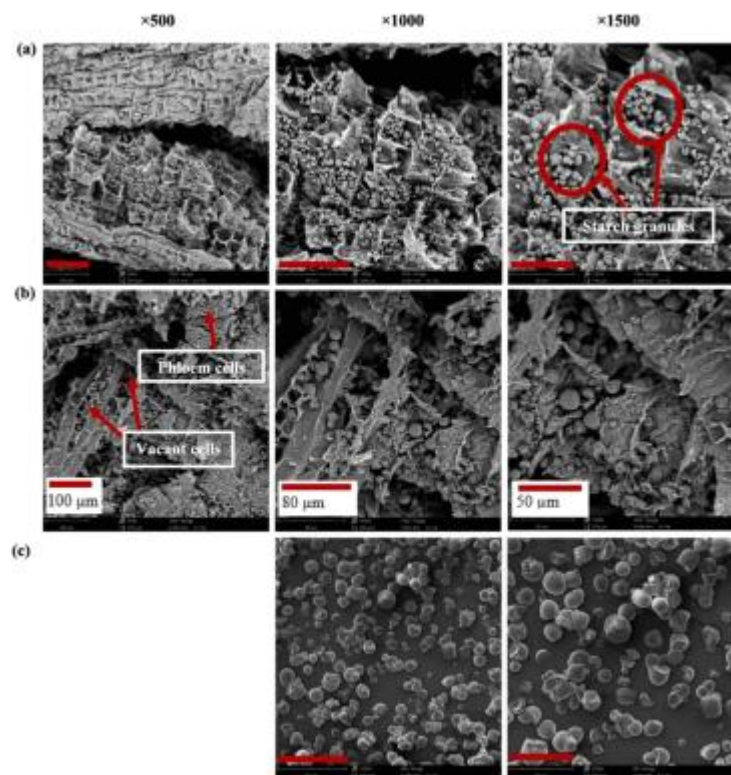
S. no.	Cellulose (%)	Hemicellulose (%)	Lignin (%)	Reference
1	13.75	37.86	9.14	29
2	5.5–15	41–65	9–16	30
3	14.17	23.40	10.88	31
4	42.20	43.80	12.40	32
5	37.90	37.00	7.50	33
6	37.90	23.90	7.50	34
7	37.90	37.00	7.52	35
8	37.9	23.90	7.5	36

Cassava peel and bagasse can be utilized in various ways to reduce waste and provide additional economic and environmental benefits. In food packaging applications, cassava bagasse-based cellulose and starch can be widely used in the preparation of packaging films. Nanocellulose prepared from cassava bagasse is also used as a filler to enhance the mechanical properties, elasticity, and transparency of cassava-based films.[37](#) These films are used in the packaging of various food items, including meat products. They also show a high activity against microorganisms.[38](#) Cassava pomace is also considered one of the good sources for packaging due to its high content of starch and dietary fibre.[39](#) Apart from

this, cassava peel is also used in animal feeding due to its high energy content, nutritional value, and availability. Besides, it is also used in fish and poultry feeding. Cassava peel is considered a good dietary replacement for maize due to its low cost. In 2022, Aro *et al.* conducted research on the microbial fermentation of cassava waste. This fermented cassava waste was then used as a dietary replacement for maize. This suggests that cassava waste can be repurposed as a feedstock or ingredient in animal feed or other agricultural applications, when the price of traditional cereal grains, such as maize, is high.⁴⁰ Cassava peel, with its composition of cellulose, hemicellulose, and lignin, can be used to prepare activated carbon. This approach is considered environmentally friendly and sustainable, as it repurposes agricultural waste (cassava peels) and reduces the need for more traditional sources of carbon materials.²⁹ In 2012, Adowei *et al.* conducted a study to compare the properties of commercial activated carbon and cassava peel-derived activated carbon. From the study, it has been observed that activated carbon derived from cassava peel exhibits similar properties to those of commercial activated carbon.⁴¹ Cassava peel is also considered a good source of organic fertilizer, which is used to enhance soil fertility.⁴² Moreover, it is also used as a feedstock for the production of bioethanol.⁴³

Another research carried out by Thuppahige *et al.* observed the surface morphology of cassava peel, bagasse and commercially available cassava starch. As represented in [Fig. 3\(a\)](#), the honeycomb-like and heterogeneous chambers of the cassava peel have been observed. Each chamber is strongly enclosed by thick cell walls composed of sclerenchyma cells. Moreover, a significant amount of starch granules was also observed within the cassava peel. These starch granules were found to be round, irregular, and closely adhered to the cell walls. The morphological structure of cassava bagasse was also shown in [Fig. 3\(b\)](#). A thin and disordered arrangement was observed in the case of cassava bagasse, which is associated with the vascular bundles. This vascular bundle is associated with the presence of fibrous structure of the cassava bagasse. Moreover, a large amount of round and irregularly shaped starch granules is also observed in cassava bagasse. Furthermore, from the scanning electron microscopy (SEM) analysis, it has also been observed that starch granules obtained from cassava bagasse and cassava peel show the same morphology as commercial cassava starch. Based on the SEM micrographs, it has been noted that the surfaces of commercial cassava starch granules exhibit a smooth texture. Additionally, the chemical composition of cassava peels and bagasse was also analysed in this research. It has been observed that both peel and bagasse contain high amounts of carbon and oxygen. The other important mineral compounds, such as Ca, Na, Fe, Al, Si, and others, were also present. The thermal properties of cassava peel and bagasse have been studied using thermogravimetric analysis. Three stages of thermal

decomposition have occurred in both cassava peel and bagasse. The initial phase was related to the loss of mass due to the evaporation of water. In the case of cassava peel, the loss in mass was around 9.26%, whereas in the case of cassava bagasse, it was 1.18%. Based on these findings, it has been concluded that cassava peel shows higher moisture activity compared to cassava bagasse. There was an additional thermal decomposition stage obtained for the peel, which occurred between 139.4 to 251.5 °C. The second decomposition phase associated with the degradation of starch was observed at 250 °C for cassava peel and 235 °C for bagasse. In the third decomposition stage, the mass loss of cassava peel was 25.28%, slightly higher than that of cassava bagasse. The higher mass loss also indicates the presence of more fibrous material in cassava peel. Therefore, it can be concluded that both cassava peel and bagasse exhibit slightly different thermal properties.[44](#)



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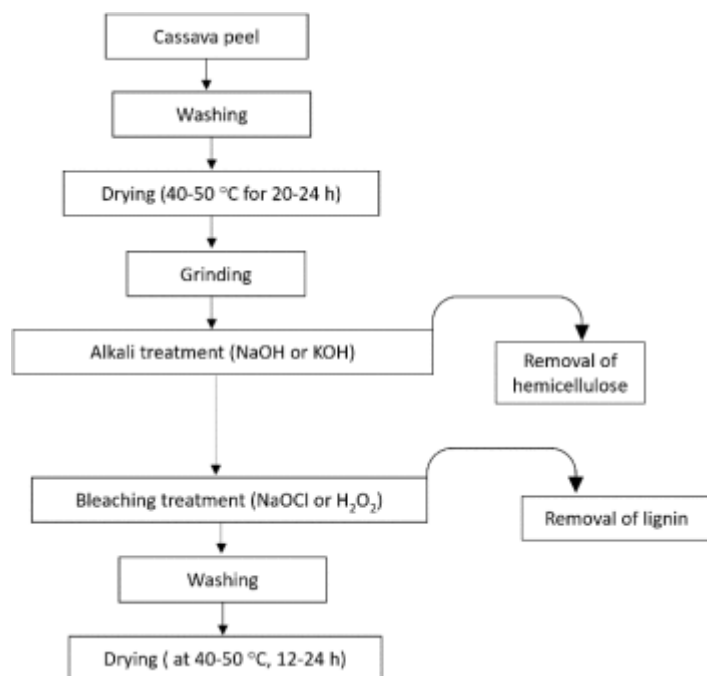
Fig. 3. Scanning electron microscope micrographs representing the surface morphology of (a) cassava peel, (b) cassava bagasse, and (c) commercial cassava starch. Reproduced from Thupphahige *et al.*, 2023.[44](#) Copyright Elsevier 2023.

Extraction of polysaccharides from cassava-based wastes

Extraction of cellulose from cassava-based wastes

Researchers and scientists have employed various extraction methods to obtain cellulose from agro-waste materials.⁴⁵ In agricultural residue, cellulose fibres consist mainly of cellulose, along with other substances such as hemicellulose, lignin, and pectin. These cellulose fibres can be predominantly isolated from waste materials through a chemical extraction process. Chemical extraction is the most commonly used treatment for the extraction of cellulosic materials. The pretreatment is an essential step in the chemical treatment of cellulose extraction, which is typically done to isolate cellulose fibre from natural sources such as wood or agricultural waste. Two classical approaches for the pretreatment of these biomasses are acid and alkali treatments.^{46, 47} The acid treatment, also referred to as the bleaching process, is used to eliminate lignin and other undesired impurities. Commonly used bleaching agents include chlorine dioxide (ClO_2), hydrogen peroxide (H_2O_2), and sodium hypochlorite (NaOCl). The occurrence of a clean, white fibre indicates the effective removal of lignin and other undesirable elements. Apart from cassava peel, several other agricultural waste materials are also subjected to pretreatment for cellulose extraction. These include orange peel, sugarcane bagasse, bamboo bagasse, rice hulls, apple stem, mulberry bark, and others.

In 2017, Widiarto *et al.* conducted a study to extract cellulose from cassava peel by chemical treatment (an alkaline pretreatment method followed by a subsequent bleaching process). The treatment was done for 4.5 h at 90 °C. This method provided a yield of 17.8% cellulose content based on the dry weight of cassava peel. In comparison, nitric and sulfuric acid pretreatment methods resulted in lower cellulose yields of about 10.78% and 10.32%, respectively.⁴⁸ In 2019, Widiarto *et al.* conducted another research for the extraction of cellulose from cassava peel, where the yield of cellulose obtained from the experiment was 17.80%.³⁶ A study was conducted by Travalini *et al.* in 2018 on cassava bagasse, where chemical treatment was applied for cellulose extraction. Initially, a bleaching treatment was applied for 5 h, followed by an alkali treatment for a duration of 1 h. The analysis revealed that the cellulose content in cassava bagasse was found to be 51.5%, with a corresponding crystallinity index of 49.3%.⁴⁹ Huang *et al.* (2020) investigated a method for processing cassava peel, starting with amylase pretreatment followed by cellulose bleaching. The process involved enzymatic hydrolysis for 3 h, followed by bleaching treatment at 70 °C for 2 h.⁵⁰ In a study, Abiazem *et al.* (2019) investigated the chemical treatment of cassava peel through alkali treatment, followed by bleaching treatment to extract cellulose. The resulting yield obtained from these processes was 6.42%, with a measured crystallinity index of 91.72%.⁵¹ Based on the above literature, the chemical extraction process of cellulose from cassava peel has been explained in the flowchart presented in [Fig. 4](#).



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Fig. 4. Extraction of cellulose from cassava peel using chemical treatment.

Extraction of nanocellulose from cassava peel

The fabrication of nanocellulose started many years ago. Nanocellulose exhibits some important properties, such as good thermal and mechanical properties, barrier properties and optical properties. The behaviour of nanocellulose makes it well-suited for food packaging applications.⁵² There are various methods for the extraction of nanocellulose, and among these, the acid hydrolysis technique is widely used. This approach is effective and rapidly yields nanocellulose with improved properties. It removes the disordered and irregular segment of cellulose and arranges it in an ordered form. The process involves several stages, including hydrolysis of cellulose using concentrated acid under controlled conditions. Then, the addition of ice-cold water stops the reaction. Further centrifugation, dialysis and ultrasonication need to be done to get the final product.³⁶ There are several studies that have been conducted to investigate nanocellulose production from cassava peel and bagasse. In 2017, Leite *et al.* conducted research on the extraction of nanocellulose from cassava peel. A solution of sulfuric acid was employed for the acid hydrolysis process, where the temperature was maintained at 60 °C for a period of 90 min. Following that, the mixture was cooled to 40 °C, and distilled water was used to dilute the suspension. The suspension was neutralized by using potassium hydroxide (KOH). After

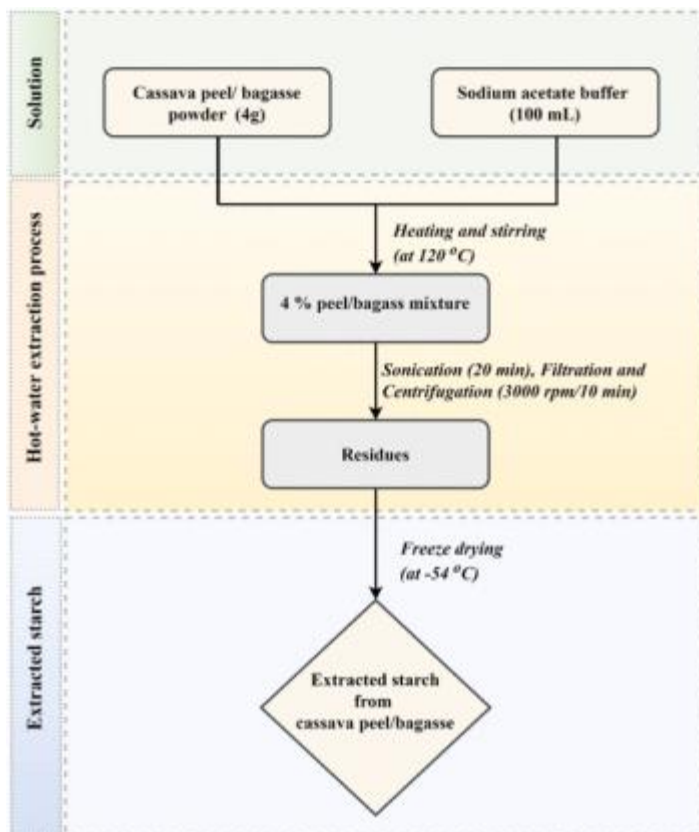
neutralization, the solution was subjected to centrifugation and homogenization to produce the nanocellulose.⁵³ Another study conducted by Widiarto *et al.* extracted the cassava peel-based nanocellulose using acid hydrolysis. Cellulose extracted from cassava peel was hydrolysed for 2 h using sulfuric acid (45% (w/w)). After that, ice-cooled distilled water was used to terminate the reaction. The resulting suspension was then subjected to centrifugation at 14 000 rpm. Dialysis was carried out over 5 days to neutralize and remove the salt present in the suspension. Finally, the suspension was sonicated and freeze-dried overnight to form the nanocellulose. The Transmission electron microscopy (TEM) analysis confirms the particle size of nanocellulose, which was below 150 nm.⁴⁸ In a research conducted by Huang *et al.* in 2018, nanocellulose was extracted from cassava residue by chemo-mechanical method. Here, phosphoric acid was used for acid hydrolysis treatment. The combined mixture of cellulose and phosphoric acid was transferred into a high-speed disperser machine operating at 800 rpm, where it was stirred at 27 °C for 2 h. After that, ice cubes were used to terminate the reaction. The resulting suspension was then subjected to centrifugation and washed with ethanol until it reached neutral pH. At last, the suspension was sonicated for 90 min at 40 W, 50 °C, and a frequency of 20 kHz, and then kept for drying. The SEM image shows a fibrous structure in the nanoscale range.⁵⁴

Extraction process of starch and its derivatives from cassava peel

One of the most widely used and inexpensive agricultural products that is completely biodegradable in many situations is starch. Starch primarily originates from tubers such as potatoes and cassava, and grains such as wheat, corn, and rice. It is the main source of energy for the plant and is found in the seeds or roots. Starch is also a polymer, which is an important aspect of its numerous applications in many industries. Cassava starch can be plasticized, strengthened with fibres, or combined with other polymers to improve its characteristics. It is also now employed as a raw material for the production of bioethanol and renewable energy. Starch can be extracted from cassava peel in a planned and purposeful manner. Till now, many studies have been done on the extraction of starch from cassava peel.

In 2023, Fronza and his co-researchers conducted research to extract starch by the polarimetric method. The method primarily comprises two steps. The first step involved hydrochloric acid treatment, followed by clarification and filtration. Afterwards, the sample was treated with 40% ethanol and re-acidified with hydrochloric acid. The optical rotation of the solution was measured with a polarimeter. The results showed that around 6.5% of starch was obtained from cassava peel through extraction. The starch obtained from cassava peels exhibited a high thermal stability and showed degradation at 300 °C.⁵⁵ In

another study, the yield of cassava starch was found to be around 30.17% on a dry basis,⁵⁶ where a hot water extraction method was followed to extract starch from cassava peel and cassava bagasse, as represented in Fig. 5. In this method, cassava peel/bagasse was mixed with a sodium acetate buffer solution, and the mixture/suspension was heated at 120 °C, followed by cooling. Afterwards, the cooled solution was sonicated and filtered, followed by washing and freeze-drying to obtain the final starch powder. Further, a study conducted by Abel *et al.* in 2021 extracted starch from cassava peel by the blending method. In the initial step, the peels were soaked in water and continuously stirred for 45 min. After this treatment, they were kept for drying at 70 °C, and starch was obtained as the final product.⁵⁷ Dasumiati *et al.* extracted the starch with the help of a grinding method. Proper mixing of the cassava peel was done for the extraction process. Around 15% amylose and 73% amylopectin were obtained from the cassava starch.⁵⁸ A study conducted by Maharsih *et al.* for the extraction of starch using various treatments reported that the extraction was carried out using blending, filtration, and settling approaches. Drying was carried out at 50 °C for 6 h to get the final product. From the Fourier-transform infrared spectroscopy (FTIR) analysis, the presence of –OH, C–C and C–H groups indicated the presence of starch.⁵⁹ In a study carried out in 2016, Siagian *et al.* discussed the utilization of cassava peel through a series of processes, including blending, filtration, and drying for the production of cassava peel-based starch.⁶⁰



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Fig. 5. Extraction of starch *via* hot water extraction using cassava peel and cassava bagasse. Reproduced from Thupphahige *et al.*, 2023.⁵⁶ Copyright Elsevier 2023.

Properties of cassava peel-derived polysaccharides for food packaging

Cellulose and nanocellulose-based films

Barrier, mechanical, thermal and other properties

Cellulose and nanocellulose are two of the most important polysaccharides obtained from cassava peel. In recent years, these materials have gained a lot of attention due to their various properties and applications. Cellulose nanomaterials exhibit excellent thermal, shear-resistant, and water and gas-barrier properties. These characteristics make them suitable for packaging applications. Apart from this, these polysaccharide-based packaging may also enhance the shelf life of food products due to their excellent antimicrobial and antioxidant properties. If we consider cassava peel-based cellulose and nanocellulose, they also show very good barrier and mechanical properties, which make them suitable as reinforcing materials like composites and films. The barrier properties of the nanocellulose based films are commonly determined using the water vapor transmission rate (WVTR). A lower WVTR value is generally desirable, as it indicates excellent resistance to moisture. The addition of cassava peel-based nanocellulose into a film matrix effectively reduces the WVTR value because of the complex path arising from the ordered crystalline parts of nanocellulose. In addition to barrier properties, the incorporation of nanocellulose also decreases the water and oil absorption properties of the composite films. This is due to the reduction of pores and voids present in the composite materials. Furthermore, nanocellulose facilitates an interfacial bonding between the polymer matrix, and thus improves the mechanical properties. Apart from this, cellulose nanomaterial is also thermally stable. The decomposition temperature typically exceeds 200 °C, and this makes them suitable for high-temperature applications. In 2020, Huang *et al.* conducted research aiming to investigate the barrier and mechanical characteristics of nanocomposite films derived from cassava starch. The research findings indicated that the incorporation of cellulose nanofibers (CNF) lead to an increase in the tensile strength of the film, and it reached the peak value at 3–4% CNF concentration. Moreover, compared to the pure cassava starch-based film, the CNF-based film showed a decrease in water vapor permeability due to the formation of a hydrogen bond between cassava starch and CNF. Besides, CNF/thermoplastic cassava starch (TPS)-based film showed a more hydrophobic nature because of the hydroxyl groups present on the surface,

compared to the pure cassava starch film.⁵⁰ Another study was conducted by Abdullah *et al.* on microcrystalline cellulose (MCC)/cassava starch-based bioplastic. From the study, it was observed that incorporating MCC enhanced the water contact angle of the bioplastic sheet, thereby increasing the hydrophobicity of the sheet. Moreover, increasing the MCC content was associated with improvement in the mechanical properties, including Young's modulus and tensile strength of the sheet. The enhancement in tensile strength was due to the adhesion occurring at the interface between MCC and starch. Moreover, there was a decrease in the elongation caused by the volume fraction and dispersion pattern of the reinforcing agent within the matrix. Apart from this, thermogravimetric analysis (TGA) revealed that incorporating MCC into bioplastics improved the heat resistance capacity of the sheet.⁶¹ Further, Huang *et al.*, in 2018, conducted research on the mechanical properties of poly(lactic acid) (PLA)/cassava residue-based nanocellulose. The study showed that the addition of nanocellulose (NC) in the 0.5% to 1% range improved the tensile strength of PLA-based nanocomposite film compared to the neat one. Further, an increase in the concentration of nanocellulose decreased the tensile strength due to the agglomeration of nanocellulose. At 1% NC concentration, the elastic modulus of the composite film was found to be the highest, due to the development of a network structure between the nanocellulose and PLA. As a result, the intermolecular force between the molecules increased, making the film more resistant to deformation.⁵⁴ In 2025, Afzia, Bora and Ghosh reported the development of a cling wrapper formulated using pectin, pullulan, olive oil and cassava peel-based CNF for chicken meat packaging. The developed films enhanced the quality attributes of chicken meat.⁶² In another study, Saravanan *et al.* (2023) investigated the incorporation of cassava tuber peel-based nanocellulose, polyester resin, and satin-weaved bamboo fibres into composite films. They found that the load-bearing capacity of the composite film increased up to 4 vol% cellulose, due to the strong interaction of cellulose with the polyester resin. Additionally, the thermal stability of the film was enhanced, with the decomposition temperature increasing at 4 vol% cassava nanocellulose compared to the pure polyester film.⁶³ Additionally, in a study by Wang *et al.* (2018), it was observed that the addition of cassava peel-based cellulose nanomaterials (CNMs) to polymer composites can significantly enhance their oxygen barrier properties. However, the researchers also noted that CNMs exhibited sensitivity towards moisture. Therefore, it is important to select the polymeric material carefully in order to reduce the sensitivity of CNMs and improve their overall performance.⁶⁴

Starch-based films

Barrier, mechanical, thermal and other properties

Starch is one of the major components obtained from agro-waste based sources. It is one of the promising candidates in food applications. It is widely used as a thickener, stabilizer, binder and gelling agent in various food items. Beyond the traditional uses, it is also an important sustainable source for the production of biocomposite films. Such films possess excellent barrier, mechanical, optical, and antibacterial properties, thus making them suitable for packaging a large number of food items like bread, fruits, vegetables, meat products, and others. In particular, cassava starch serves as an excellent source for the advancement of nanoparticles. Films prepared from starch are considered biodegradable, renewable and non-toxic.⁶⁵ Costa *et al.*, in 2017, conducted research to assess the impact of nanocrystalline starch on cassava starch-based film. The study highlighted that the addition of nanocrystals enhanced the tensile properties and reduced the water vapor permeability (WVP) of the film.⁶⁶ Additionally, Souza *et al.* (2012) conducted a study to investigate the change in barrier and mechanical properties of composite films based on cassava starch, with the addition of starch nanomaterials and glycerol. The presence of glycerol significantly affected the tensile strength of the film. There was also an enhancement in the barrier properties of the film. Moreover, the combined effect of glycerol and nanoparticles remarkably enhanced the tensile strength and barrier properties.⁶⁵ Another study conducted by Pelissari *et al.* investigated the barrier, mechanical and antimicrobial properties of cassava starch/chitosan/oregano essential oil (OEO)-based film. The study revealed that cassava starch-based film showed the maximum WVP value. However, when chitosan was mixed with cassava starch, it showed a reduction in WVP. This is mainly due to the hydrogen bond formed between the hydroxyl (OH⁻) group of cassava starch and the amine (NH₂) group of chitosan. The addition of OEO also decreased the WVP of the film. Moreover, there was no significant change in the film with the incorporation of chitosan and OEO.⁶⁷ Additionally, Tongdeesoontorn *et al.* conducted research on the physical properties of cassava starch/carboxymethyl cellulose (CMC)-based biodegradable film. The properties of the film were improved with the addition of CMC and glycerol. There was an increase in water solubility and tensile strength of the film, and a decrease in elongation with the addition of CMC. Moreover, differential scanning calorimetry (DSC) analysis proved the uniformity of the film.⁶⁸ Research was conducted by Jumaidin *et al.* to understand the change in thermal, biodegradable and mechanical properties of thermoplastic cassava starch (TPCS)-based film with the addition of durian peel fibre (DPF). The study revealed that there was a change in the mechanical and thermal stability of the film when different concentrations of DPF were incorporated into the film. Furthermore, FTIR spectroscopy and SEM showed excellent interactions between DPF and the TPCS matrix.⁶⁹ Luchese *et al.* conducted another research on cassava starch/blueberry pomace composite film. The research mainly focused on assessing the effect of blueberry pomace on cassava starch film. From

the DSC analysis, it was observed that the addition of blueberry pomace significantly improved the thermal stability of the film compared to the film without blueberry pomace. This may be due to the interaction between the blueberry pomace and cassava starch. The addition of blueberry pomace may also increase the UV light protection capacity of the film. Apart from this, the swelling behaviour was also found to be increased.⁷⁰ Some of the barrier, mechanical and thermal properties of cellulose- and nanocellulose-derived from cassava peel are represented in [Table 2](#).

Table 2. Barrier, mechanical and thermal properties of cellulose and nanocellulose derived from cassava peel for packaging applications^a

S. no.	Sources	Properties	References
1	•Thermoplastic cassava starch (TCS)	•Tensile strength of the film reaches the peak value at 3–4% CNF concentration	50
	•Glycerol	•Water vapor permeability of the CNF-based film decreases compared to pure cassava starch-based film	
	•Cassava peel-based-CNF	•CNF/TCS based film shows a more hydrophobic nature	
2	•Cassava starch	•Hydrophobicity of the bioplastic sheet increases	61
	•Glycerol	•Reduces the moisture uptake of the bioplastic sheet.	
	•Microcrystalline cellulose (MCC)	•Young's modulus and tensile strength of the sheet increase with an increase in the MCC. •Elongation decreases with an increase in the concentration of MCC.	
3	•PLA	•Addition of NC in the 0.5% to 1% range increases the tensile strength of PLA-based nanocomposite film. Further increase in nanocellulose concentration decreases the tensile strength	62

S. no.	Sources	Properties	References
	•Cassava peel-based nanocellulose	•CNF (1–1.5 wt%) improved the mechanical, thermal, and barrier properties of the pullulan/pectin-based films	
	•Pectin		
	•Pullulan		
	•Olive oil		
4	•Cassava tuber peel-based nanocellulose	•Load-bearing capacity of the composite film increases up to 4 vol% cellulose due to the strong interaction of cellulose with the polyester resin	63
	•Polyester resin	•The thermal stability of the film was found to be increased. At 4 vol% cassava nanocellulose, the decomposition temperature increases compared to pure polyester film	
	•Satin-weaved bamboo fibre		
5	•Cassava peel-based cellulose nanomaterials	•Addition of CNMs to polymer composites can enhance their oxygen barrier properties	64
	•Polymer	•CNMs are sensitive to moisture; they may require additional design considerations, such as multilayered structures and the choice of surrounding polymers, to address their moisture sensitivity and improve their overall performance	
6	•Cassava peel-based starch	•Thickness of the film increases	65
	•Glycerol	•Water vapor permeability of the film decreases	

S. no.	Sources	Properties	References
	•Starch nanoparticles	•No significant change in the tensile strength of the film compared to the control	
7	•Chitosan	•Barrier properties of the film improved	67
	•Cassava starch	•Antioxidant properties of the film increase	
	•Oregano essential oil	•Increase in the flexibility of the film	
		•Reduction in the WVP of the film	
		•No significant change in the thermal stability	
8	• Cassava peel starch	•Decrease in the water solubility of the film with the addition of CMC.	68
	•CMC	•CMC incorporated film becomes more homogenous	
	•Glycerol	•Increase in the tensile strength of the film	
		•Decrease in the elongation	
9	•Durian peel fibre (DPF)	•Improvement in the thermal stability of the film	69
	•Thermoplastic cassava starch (TPCS)	•Increase in the rate of biodegradability of the film	
		•Flexural and tensile strength of the film improved	
10	• Cassava starch	• Enhanced the barrier properties of the film	70
	•Blueberry pomace	•Improvement in the thermal stability of the film	

S. no.	Sources	Properties	References
	•Sorbitol	•High UV light barrier properties of the composite film •Swelling capacity of the film slightly increases with the addition of blueberry pomace	

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TCS: thermoplastic cassava starch; CNF: cellulose nanofibers; MCC: microcrystalline cellulose; PLA: poly(lactic acid); NC: nanocellulose; BHT: butylated hydroxytoluene; CNMs: cellulose nanomaterials; WVP: water vapor permeability; CMC: carboxymethyl cellulose; DPF: durian peel fibre; TPCS: thermoplastic cassava starch; FTIR: Fourier-transform infrared spectroscopy; SEM: scanning electron microscopy; UV: ultra violet.

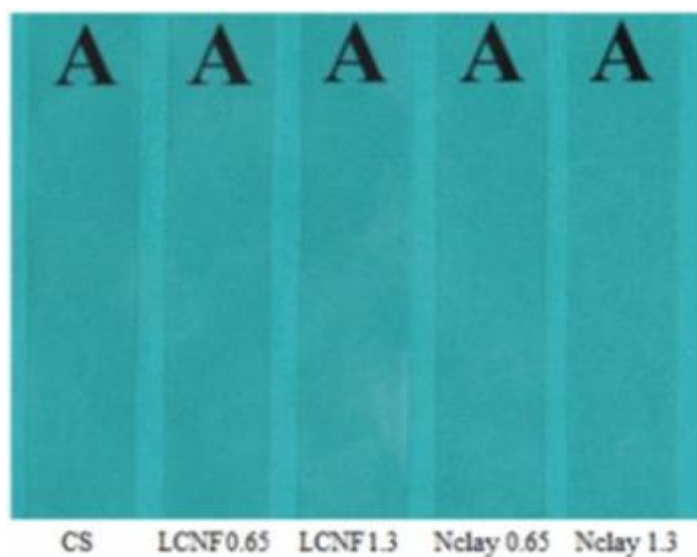
Cassava and cassava peel-derived polysaccharides for food packaging applications

Cellulose is considered as one of the most remarkable and versatile natural substances that has gained increasing attention for its eco-friendly and biodegradable properties. It is the most abundant organic polymer on Earth. Cellulose is anticipated to be non-toxic and safe for food contact applications. Moreover, cellulose-based materials can have an impressive strength-to-weight ratio, which makes them suitable for various packaging applications. They can provide the necessary strength and durability while keeping packaging lightweight, reducing shipping costs and energy consumption.⁷¹ Furthermore, the structure of cellulose imparts characteristics necessary for bioengineering uses, including biocompatibility, biodegradability, and suitable mechanical attributes.⁷² Despite all the properties listed above, it has a limited range of applications. One potential solution for this issue involves utilizing nanomaterials as a filler in the production of composite materials.⁷³

Cellulose nanoparticles indeed offer a promising approach for the development of sustainable packaging materials with enhanced properties. When cellulose is reduced in size from micro to nanoscale, several changes may take place, which can significantly impact the utility of these nanoparticles in the field of sustainable packaging.⁷⁴ The development of cassava peel-based cellulose and nanocellulose for packaging is still a growing field, and research is ongoing to optimize extraction processes and improve the performance of these materials. However, they hold great promise in addressing the environmental concerns associated with conventional packaging materials. Cellulose derived from cassava peel can be used to produce films and coatings for various packaging

applications. These materials can be used for food packaging, as well as for non-food items such as electronics and textiles. The biodegradable bags made from cassava peel-based cellulose can reduce plastic pollution and have a lower environmental impact. Moreover, combining cassava peel-based nanocellulose with other biodegradable polymers can result in stronger, more sustainable packaging materials. This type of packaging can help extend the shelf life of food products by providing effective barrier properties and protection against environmental factors.

Cassava-based starch in sustainable packaging is an innovative and eco-friendly approach that can replace the plastic-based packaging and promote environmentally friendly practices in the packaging industry. It offers the advantages of biodegradability, renewability, and customization, making it a promising option for reducing the environmental impact of packaging materials. To address the global concern over plastic bags, cassava biopolymers have been introduced into the market. These bags are crafted from starch derived from cassava plants, making them biodegradable and eco-friendly. Moreover, these bags decompose faster than petroleum-based plastic bags, making them an eco-conscious choice.⁷⁵ In this context, the visual appearance of films based on neat cassava starch, cassava starch with lignocellulosic nanofibers (LCNFs), and nanoclay (Nclay) is shown in [Fig. 6](#), where the films have possible food packaging applications due to their tensile and barrier properties.⁷⁶ The films based on cassava starch reinforced with LCNF have better properties in comparison to biocomposites of cassava starch and Nclay. A study conducted by Kaisangsri *et al.* focused on the development of foam trays from cassava starch blended with chitosan and kraft fibre. The study found that the development of foam trays has the potential to replace traditional polystyrene foam with a more environmentally friendly and sustainable alternative. Moreover, cassava starch-based foam displayed improved properties compared to polystyrene foam in terms of density, tensile strength, and elongation.⁷⁷ Wahyuningtiyas and Suryanto conducted a study to examine the impact of incorporating nanoclay into cassava starch along with glycerol as a plasticizer, to produce bioplastic. The research indicated that the addition of nanoclay significantly enhanced the properties of cassava starch-based bioplastics. Enhancement in tensile strength, elongation, and water absorption, along with the material's biodegradability, making it a promising option for environmentally friendly packaging and other applications where traditional plastics are commonly used. This research contributes to the development of sustainable materials with reduced environmental impact.⁷⁸ Furthermore, some of the other applications of cassava-based starch have been represented in [Table 3](#). In this regard, cassava and cassava peel-based biopolymers have received considerable attention in developing films for packaging applications.



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Fig. 6. Visual appearance of films based on cassava starch (CS), CS with lignocellulosic nanofibers (LCNF) and nanoclay (Nclay). Reproduced from Travalini *et al.*, 2019.[76](#) Copyright Elsevier 2019.

Table 3. Applications of cassava starch in sustainable packaging[a](#)

S. no.	Materials	Methods of application	Food packaging/coating applications	Effect on the shelf life of food products	References
1	Cassava starch, sucrose, invert sugar, glycerol	Solvent casting	Film is used for packing toast	—	65
2	Cassava starch, glycerol, glutaraldehyde, polyethylene glycol	Solvent casting	Used as an edible film	—	79
3	Cassava starch, grape stalks, guar gum,	Hydraulic pressing	The foam is used for storing English cake	—	80

S. no.	Materials	Methods of application	Food packaging/coating applications	Effect on the shelf life of food products	References
	magnesium stearate, glycerol				
4	Cassava starch, zinc nanoparticles, glycerol	Solvent casting	The nanocomposite is used for the packaging of tomatoes	Films with 1–2% zinc nanoparticles suppressed the microbial growth and better retained tomato quality than LDPE at 27 °C on day 9	81
5	Cassava starch, chitosan, sorbitol, citric acid, gelatin	Solvent casting	Used in the coating of fruits	—	82
6	Cassava starch, citric acid, glycerol	Dipping	Used for coating mango slices	- The respiration rate of coated mango slices was reduced by 41% compared to the control - Texture and color characteristics of the coated mango slices were also found to be maintained during storage - Coated mango slices were well	83

S. no.	Materials	Methods of application	Food packaging/coating applications	Effect on the shelf life of food products	References
				accepted by consumers	
				Coating significantly extended the shelf life of the fresh-cut mangoes	
7	Cassava starch, glycerol, <i>Kaempferia rotunda</i> , essential oil of <i>Curcuma xanthorrhiza</i>	Dipping	Used for coating patin (fish) fillets	Edible coating prepared with cassava starch, <i>Kaempferia rotunda</i> and essential oil of <i>Curcuma xanthorrhiza</i> extended the shelf life of patin (fish) fillet.	84
8	Cassava starch, sodium-CMC, LAB	Solvent casting	Used in the packaging of bananas	A banana wrapped with the composite film exhibited delayed browning and black spot development compared with the unwrapped banana. Thus, the film enhanced the freshness and shelf life	85

S. no.	Materials	Methods of application	Food packaging/coating applications	Effect on the shelf life of food products	References
9	Cassava peel starch, sorbitol, MCC	Solvent casting	Used for the preparation of bioplastics	—	57
10	Cassava peel-based starch, glycerol, kaffir lime essential oil, citric acid	Solvent casting	Used for the production of bioplastics		86
11	Gelatin powder, AgNPs from cassava peel, glycerol	Solvent casting	The film produced is used for the preservation of sapodilla fruits	Coated sapodilla fruits exhibited reduced weight loss and microbial growth, thus enhancing the quality and shelf life of sapodilla fruits	87
12	Cassava starch, fructose, cassava bagasse, cassava peel	Solvent casting	Used for the preparation of films	—	9

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CMC: carboxymethyl cellulose; LAB: lactic acid bacteria; MCC: microcrystalline cellulose; AgNPs: silver nanoparticles.

Food-based applications of cassava peel

The valorization of agro-industrial residues has become increasingly important as the demand for energy and value-added bioproducts continues to rise. Among these residues, cassava processing generates large amounts of peels that are often overlooked despite their promising potential in food and feed applications. Cassava peels are rich in xylan, making them an excellent source for producing xylo-oligosaccharides (XOS), prebiotics

known for their beneficial effects on gut health.⁸⁸ Beyond prebiotic extraction, the peels can be transformed into light, crispy snacks that are not only palatable but also carry market value. In addition to human consumption, cassava peels retain considerable amounts of carbohydrates and dietary fiber, enabling their inclusion in animal diets as an effective energy source.⁸⁹ For instance, studies on *Clarias gariepinus* (African catfish) have demonstrated the potential of cassava peel as a feed component without compromising growth.⁹⁰ Similarly, in poultry nutrition, broiler chickens fed diets where maize was partially replaced with up to 25% raw or processed cassava peel meal showed not only good performance but also enhanced carcass characteristics.⁹¹

Life cycle assessment of cassava and its byproducts

Cassava and its byproducts, including peels, stems, straw, bagasse, leaves, and wastewater, have diverse applications worldwide, such as starch production, animal feed, medicinal and ornamental uses, soil regeneration, bioethanol, biofertilizer, bioplastic, and biofuel production within a biorefinery framework. These valorization pathways not only reduce competition with food security but also generate employment for unskilled labor. To ensure their sustainability, it is essential to apply novel analytical tools, such as Life Cycle Assessment (LCA), which enable stakeholders to evaluate socioeconomic and environmental impacts and make informed decisions for biorefinery projects.⁹²

In this context, the integration of cassava waste into a waste-water-energy-food (WWEF) nexus system has been proposed for small-scale cassava industries in Brazil. This system combines anaerobic digestion (AD) with co-generation or combined heat and power (CHP) plants to convert organic waste into biogas, a promising renewable energy source. Using LCA, three scenarios were compared: business-as-usual, improved business-as-usual, and WWEF closed-loop. The functional unit was defined as the production of 1 kg of cassava starch/flour, with selected impact categories including global warming potential (GWP), cumulative energy demand (CED), freshwater eutrophication potential (FEP), terrestrial acidification potential (TAP), and water depletion potential (WDP).

The results indicated that landfilling cassava waste, high power consumption during starch/flour production, and fertilizer-related emissions are major environmental hotspots under the business-as-usual case. In contrast, the WWEF closed-loop scenario demonstrated the most favorable outcomes, achieving over 90% reduction in GWP and more than 50% savings in other impact categories (except FEP with <10% savings). Sensitivity analysis confirmed the robustness of these findings.⁹³ Overall, the study highlights that efficient utilization of cassava byproducts through an integrated WWEF closed-loop system not only minimizes biological waste disposal but also reduces

dependency on fossil-based resources, enhances resource circularity, and strengthens the sustainability of rural cassava value chains.

Conclusion and future prospects

Several innovative research solutions are being applied to cassava waste biomass, which can provide advancement for developing a green environment, sustainability, renewability, and efficiency. This could also enhance the supply of biomaterials for industry and promote the adoption of a circular model for production, consumption, and disposal. Sustainable and renewable resources carry significant prospects for the well-being of upcoming generations. Thus, biodegradable and sustainable materials hold great potential for future generations. The severe environmental issues created by plastic materials can be significantly reduced by the widespread use of polysaccharides as a reinforcing agent during the preparation of composite films. The advancement of these sustainable composites will have a positive impact on the environment, reduce the need for recycling plastic waste, and lower the carbon footprint associated with petroleum-derived materials. As potential alternatives to non-biodegradable plastic products, an increasing demand for natural resources is being utilized to promote environmental sustainability. Moreover, the abundance and cost-effectiveness of these materials have received a lot of attention in recent years.

The demand for these agricultural industrial wastes has risen dramatically due to their ability to meet higher environmental, economic and social standards. Moreover, they have gained popularity because of their biodegradability, low manufacturing cost, renewability, low energy consumption, low density, as well as abundance in nature. In the food industry, cassava peel has been used for various purposes. It can be processed into cassava flour, which can be used as a gluten-free flour alternative in baking and cooking. It can also be used to make snack products like cassava chips or crisps. For biofuel production, such as biogas or bioethanol, it is used as a feedstock. Additionally, cassava polysaccharides exhibit excellent tensile strength and impressive flexural strength, as confirmed by numerous mechanical tests and research studies, making them suitable for a wide range of applications, such as packaging, lightweight constructions and industrial applications. Cassava peel shows great promise as a reinforcement for sustainable polymer composites. The utilization of cassava peel-based polysaccharides in sustainable packaging helps to reduce the harmful effects of synthetic polymers and decrease the dependence on petroleum-based products. This will not only enhance the socio-economic benefits of rural communities by increasing tax revenues and creating jobs, but also provide a significant opportunity in a variety of potential industrial applications. Further, in terms of research, more advanced technologies are required for the advancement of this

type of packaging system. This represents a novel area of research and an innovation which addresses some specific challenges for the future industrial utilization of cassava peel-based polysaccharides and their applications in the packaging industry.

Ethical approval

Compliance with ethical approval.

Author contributions

Nurin Afzia: original draft writing, review and editing. Tabli Ghosh: conceptualization, supervision, validation, final review and editing.

Conflicts of interest

The authors declare no conflict of interest.

Data availability

No datasets were generated or analysed during the study.

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